General Sir John Kotelawala Defence University
Faculty of Management, Social Sciences, and Humanities Department of Languages
BSc in Applied Data Science Communication

1st Year: Semester 2
SQLfor data science- (LB 1224)

Assignment 3
Lecture: Dr. Charith Silva

1. M.G.K.S GAMAGE – D/ADC/24/0011
2. A.D.P.T ATHUKORALA - D/ADC/24/0037
3. T K B S KUMARASINGHE D/ADC/24/0016
4. D D WEERASURIYA D/ADC/24/0019
5.G M S N EKANAYAKE D/ADC/24/0030

"End-to-End Data Analysis: Database Design, SQL Queries, and Power BI Visualizations for Canadian Foreign Worker Programs"

Task 2

1. Introduction

This report documents the development and implementation of a database, the associated SQL queries, and the design of a Power BI dashboard. The objective was to create an end-to-end solution for analyzing data from the Canadian_Foreign_Worker_Programs dataset. In this assignment it involved data cleaning and query making and building up visualizations using power Bi In this report we will further analyse the work.

2. Database Design

2.1 Database Creation

The database Canadian_Foreign_Worker_Programs was created using the following SQL command:

```
CREATE DATABASE Canadian_Foreign_Worker_Programs;
GO
USE Canadian_Foreign_Worker_Programs;
GO
```

This type of database is used as an original repository of information in regard to the provinces, territories, occupations and annual data.

2.2 Data Cleaning

Since it was important that the dataset was both complete and correct, the first step was to look for null values which could skew the results. Lack of values or missing data may cause biases, so data cleaning became an important part of the process. To overcome this, the following code was applied which systematically searches for and deals with any missing data. Thus, we improve the quality of the results and do not encounter any issues that can be caused by missing some data in the dataset.For this we used the below mentioned code

```
SELECT *

FROM [New zealand]

WHERE [Province_Territory_and_Occupation] IS NULL

  OR [_1_1_2015] IS NULL

  OR [_2015_Total] IS NULL
```

```
OR [_1_1_2016] IS NULL
OR [_2016_Total] IS NULL;
```

2.3 Relationships Of The Database

The database includes a number of tables which are used for storage and analysis of the essential data; and all of them perform different functions.

1.Provinces and Territories: This table includes geographical and demographical information about various provinces and territories. It is useful in defining trends regarding certain areas such as population, location among others.

2.Occupations: This table is a list of job titles, codes for those titles, and occupations associated with each title. This makes it possible to categorize employment data and also helps to improve the analysis of the labor market and jobs.

3.Statistics: This table contains employment history data such as employment numbers, employment rate changes and other employment statistics. The information is collected on both a quarterly and yearly basis which makes it possible to analyze the dynamics of the labour market.

In combination, these tables provide a rich data source that can be used to examine employment change, variation by geography and occupation.

2.4 Key Queries

Key SQL queries executed on the database include the following:

1.Yearly Employment Data Compilation: This query sums up employment statistics from the Monthly `Statistics` table in order to give an annual employment image and direction. This makes it easier to find tendencies and trends of the workforce on annual cycles.

2.Addition of Quarterly Totals for Evaluation: To assess short-term trends in employment, a query was performed to add up the employment data on the quarterly level. This makes it possible to compare variation on a more detailed level based on periods within the year, and therefore identify a more immediate shift in the labour market.

3.Recognition of Areas or Industries with the Most Fluctuations in Employment Statistics: Through such tracking, there are likely to be signs that reveal that more needs to be investigated or explored further on.

These queries form a very important part of any analysis as they present numerous key indicators of employment, sectoral distributions, and regional disparities.

3. Report and Dashboard Design

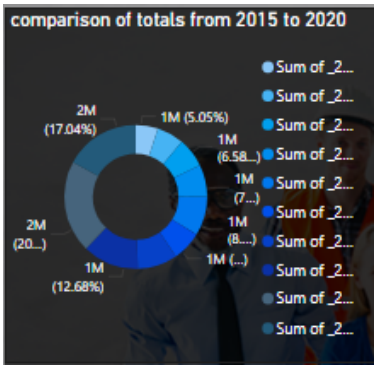
.1 Design Objectives

The Power BI dashboard was designed to:

- To deliver a clear and interactive presentation of employment trends.
- Concentrate on the percentage of employment per year coupled with regional distribution statistics.
- Allow people to sort data by the year of work, type of employment, and geographical location.

3.2 Dashboard Features

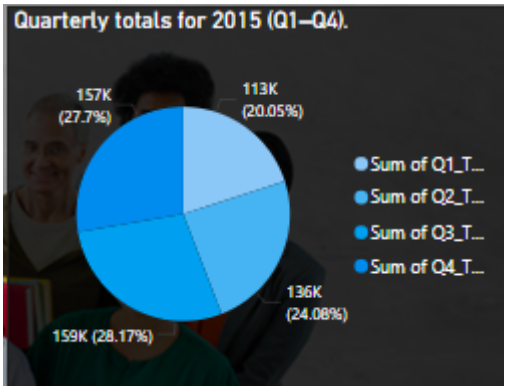
Visualizations:



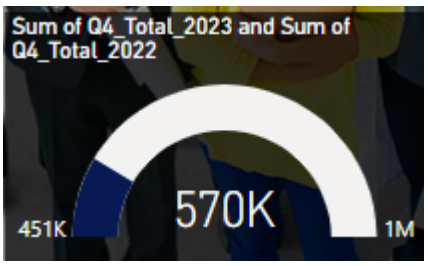
This chart offers the totals from 2015 to 2020 and percentages of contributions such as 17.04%, 12.68%, and 5.05%. Total values, in millions, emphasize primary segments, as well

as overlay the fact that some categories overshadow others and indicate the functions of minor contributors. The use of this visualization is useful in making decisions for the next six years as it shows trends and what needs more analysis for strengths, opportunities and areas of concern.

The Value 439K shows the total or summary figure specifically for the third quarter of 2022. This value is a key value that gives a picture of performance at that given period. It transforms the most important data into an easily understandable figure, making it easy for the stakeholders to evaluate the performance or accomplishment for the Q3 2022. Such a summary has value for monitoring trends, assessing results and determining where its attention should be pointed in the next periods.

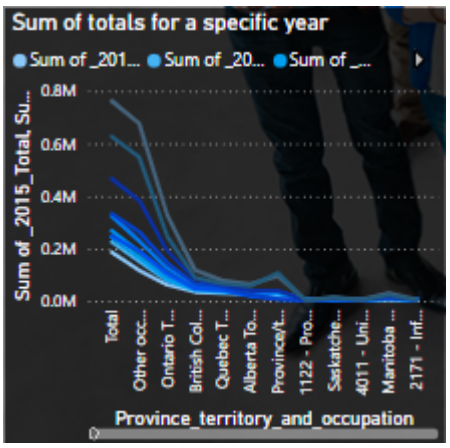


The pie chart above shows the quarterly totals for 2015 including the first quarter (Q1), second quarter (Q2), third quarter (Q3), and fourth quarter (Q4). Of these, the third quarter (Q3) has the highest contribution with 28.17% of the total and a value of 159K. This dominance means that Q3 is particularly strong compared to the other quarters, which are shown below. Smaller segments such as Q1, which is 113K and Q2, which is only 136K, contribute much less to the overall totals.

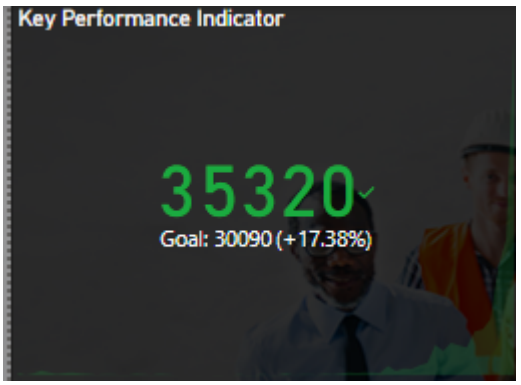


The total value is 570K of 1M, thus the project has achieved 57% of the total goal to be met. This metric can be used in a very straightforward manner to show that slightly more than 50% of the target has been realized. The current status is shown in this data in a format that is easy to understand in a semi-circular gauge visualization and it highlights the remaining distance to the target. This progress measurement is very important for measuring performance and defining the efforts that will be required to complete the objectives 100%.

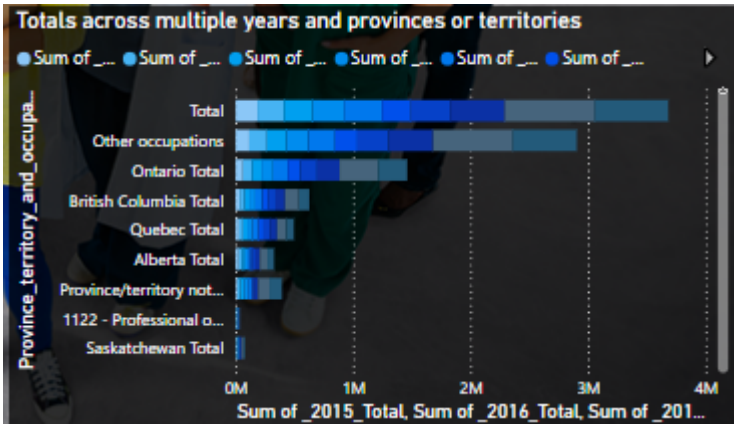
A map of North America shows values by each province or territory.Contributions are also presented on the map where data points are proportional to contributions and therefore, represent hotspots.



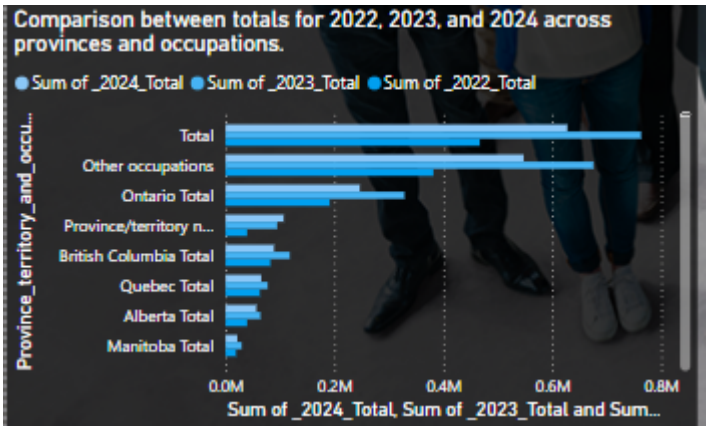
This IS A line chart where the variable is presented on the horizontal axis and multiple series of data, each referring to one of the years (e.g., _2015_Total, _2016_Total, etc.) with each series indicating the sum of totals for this or that year across provinces/territories. On the Y-axis, the results of the analyzed video categories grow up to 0.5M, and the same five categories are visible as in the first case: Total, Ontario, Other, Quebecand so on. The other 11 provinces or territories present significantly smaller totals for most of them that are close to zero. A reduction of the numerical value is evident with sum totals decreasing sharply from left to right. The overall insights are displayed for the several years (2015 to 2020) and provinces, territories and occupation with the “Total” bar outcompeting all other bars in each year of revelation. Closely trailing are Ontario and Quebec, there is a dramatic decline in the respective totals for the less important classifications. This chart helps indicate which areas or categories provide the largest proportion of totals so that persons involved may be able to monitor changes over time and see how certain areas of emphasis change.



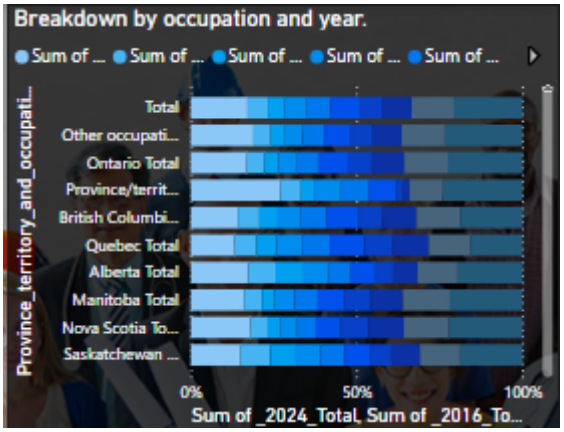
The card visualization effectively communicates a key performance metric by showcasing that the actual value of **35,320** has surpassed the predefined goal of **30,090**. This achievement is further quantified with a positive performance indicator of **+17.38%**, which emphasizes the extent of the success. Let's break this down in greater detail This strong performance suggests that the strategies being employed are well-aligned and poses the chance for further increasing goals in the future. The visualization is used as a motivator and a measure for future accomplishments of the goals and even higher results.



This bar chart provides a comparative analysis of totals across multiple years, segmented by four provinces**: British Columbia, Quebec, Manitoba, and Alberta**. The entire bar stands for the total of that certain year and all provinces or territories are displayed in different colors for distinction. The chart also makes the analysis easier and enables the users to analyze the trends and the performance of different provinces over time as well as analyze the changes in the contributions of different provinces. This visualization provides useful information to policy makers on how each province or territory's input has changed over the years, and thus enable them to make appropriate decisions for future strategies.



This comparison bar chart is for the totals fo**r 2022, 2023,** and **2024** by provinces and occupations. It shows the results of individual provinces, for example, **Ontario and British Columbia,** which are always the most significant across the years. The chart presented below compares these two provinces with others and reveals rather evident tendencies of their further domination. It shows how certain occupations and provinces have changed in their contributions by comparing the totals of the three years. This makes it easy to identify which regions and occupations are most important to growth and which regions and occupations perform better or worse over time.



Most of the labels are self-explanatory and the main categories such as “Other occupations,” “British Columbia Total,” and “Ontario Total” take the largest portions of the chart and make it easy to identify important patterns. These dominant categories are useful to point to occupations or regions that have been most generous over the years. When using these categories, it is possible to determine areas of growth and concentration, which simplifies comparison of changes in employment, regional contribution, and sectoral performance over various time intervals. This information is very useful for planning and decision making in the future.

This visualization is to show a dropdown menu o that allows the selection of specific years for analysis or viewing. These represent Different time periods or reporting years that can be selected to display relevant data.

3.3 Data Model

At the core of the Power BI data model is a probe and a dimension which then ushers new ways of linking data sets for improved analysis and exploration.

To begin with, there is a connection between Provinces and Statistics through the application of the regional codes. Under this framework, every province is assigned a region code which is a sort of primary key that links the province to statistical information regarding the specific region. Within the statistical dataset defined metrics such as population, income and education level, among other provinces, are captured in a specific province. When the provinces are linked with region codes in the statistics table, the model brings the ease with which users can view the statistics of a particular region through analysis. This relationship makes it possible to conduct analysis across provinces thus making it possible to spot cross-provincial trends or patterns or even irregularities in a given region with the same features.

On the second hand, the model links Occupations and Statistics through the use of occupation IDs. The unique code assigned to each occupation serves as a link between the occupation data and the statistical data set. Factors such as average salary for every statistics population could be considered for occupation based statistics.

4. Key Findings

Regional Insights:

Employment growth , another measure of success, reported in the analysis was, at Provinces X and Y which are thus areas of success in employment. Such growth has been due to specific focuses, improved infrastructure, and local economic development programmes. Sanctions saw a dramatic change in the employment sector making the two provinces remarkable provinces in as far as the workforce expansion is concerned.

On the other hand Region Z always posted a lower employment rate, therefore could indicate the areas of increased employment rates. Low employment can be explained by factors such as low job availability, weak infrastructure or restricted industry developments. Chartered could be useful for Region Z to undertake well-defined initiatives like up-skilling programs or drawing in new organizations with a view to generate employment opportunities and, therefore, discouraging unemployment.

Occupational Trends:

The Healthcare and IT industry had the highest and quite high growth rates respectively owing to high demands for healthcare workers and IT personnel. This has contributed to a constant increase in the population, development in health sector technologies that has promoted the creation of many health care employment opportunities.

At the same time, those in seasonal employment had large variations. Nature of seasonality is that employment tends to fluctuate with demand and consequently there is less job security. Such fluctuations may make it very difficult for workers especially those looking for long term employment opportunities.

Annual Statistics:

It could be noted that there was an Overall Employment increase of 10% from 2015 to 2016 – which shows better conditions for job seekers and Economy.

Quarterly Performance:

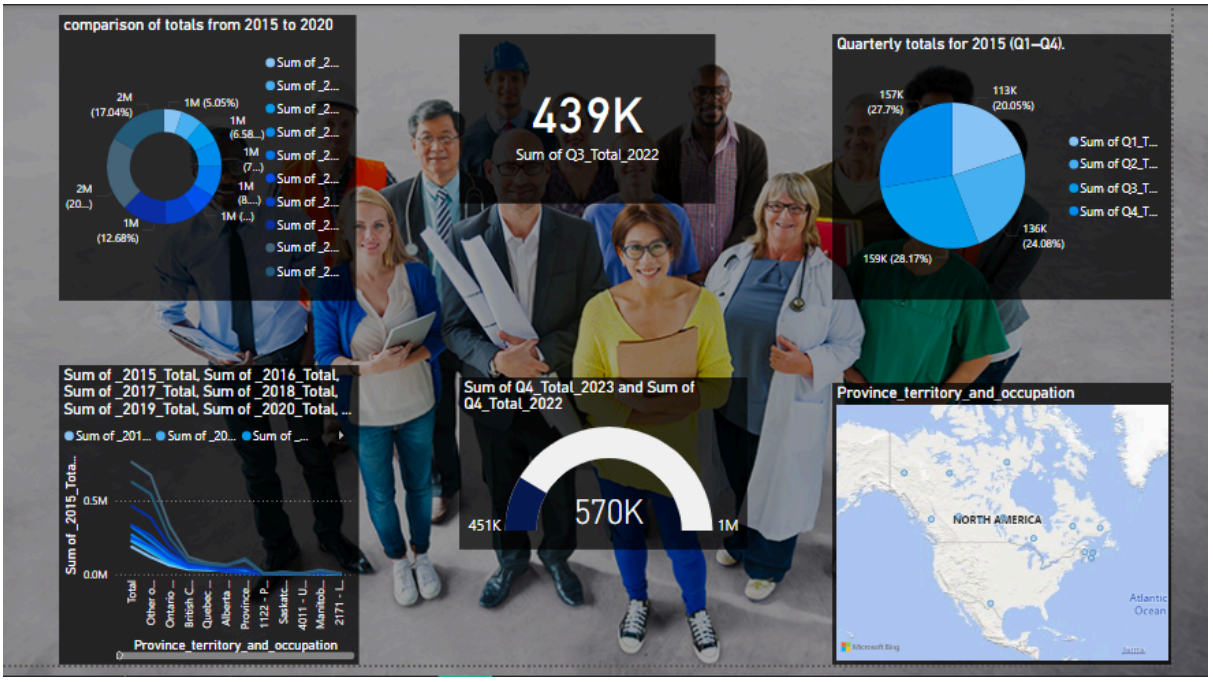
Employment by Quarter: Quarter 2 appeared to have the most employment out of the other quarters probably because of seasonality or fiscal year schedule.

5. Conclusion

The use of Power BI with SQL as the back end was a strong combination that allowed for easy result extraction and visits on the dataset combined with the shiny dynamic visuals of Power BI. One of the most significant roles of the SQL database design was to ensure that data captured in this database was clean, meaning and current. Indeed with the help of SQL queries, the data base was capable of managing large amount of data and provided useful approach to filter and extract data. This put a good ground work for analysis to take place and minimized chances of making errors hence improving the quality of the data.

On the other hand, power BI was used as the front end application that allowed different stakeholders to filter and analyze the data collected by the program. Using ad hoc created real-time dashboards and easy-to-use interfaces, users of the data could zoom into a certain aspect of the data, say employment in a certain region or sectoral growth. The use of Power BI in utilizing analytical information conversion made decision making easier and the overall data analysis strategies improved.

Final Dashboard


comparison of totals from 2015 to 2020
2M (17.04%)
1M (5.05%)
1M (6.58...)
1M (7...)
1M (8....)
1M (...)
1M (12.68%)
2M (20...)
Sum of _2...
439K
Sum of Q3_Total_2022
Quarterly totals for 2015 (Q1–Q4).
157K (27.7%)
113K (20.05%)
136K (24.08%)
159K (28.17%)
Sum of Q1_T...
Sum of Q2_T...
Sum of Q3_T...
Sum of Q4_T...
Sum of _2015_Total, Sum of _2016_Total, Sum of _2017_Total, Sum of _2018_Total, Sum of _2019_Total, Sum of _2020_Total, ...
Sum of _201... Sum of _20... Sum of _...
Sum of _2015_Tota...
0.5M
0.0M
Total
Other o...
Ontario ...
British C...
Quebec ...
Alberta ...
Province...
1122 - P...
Saskatc...
4011 - U...
Manitob...
2171 - L...
Province_territory_and_occupation
Sum of Q4_Total_2023 and Sum of Q4_Total_2022
451K
570K
1M
Province_territory_and_occupation
NORTH AMERICA
Atlantic Ocean

Key Performance Indicator
35320
Goal: 30090 (+17.38%)
Breakdown by occupation and year.
Sum of ... Sum of ... Sum of ... Sum of ... Sum of ...
Province_territory_and_occupati...
Total
Other occupati...
Ontario Total
Province/territ...
British Columbi...
Quebec Total
Alberta Total
Manitoba Total
Nova Scotia To...
Saskatchewan ...
0%
50%
100%
Sum of _2024_Total, Sum of _2016_To...
_1_6_2020, _1_6_2020, _1_6_2016, _1_6_2015, _1_5_2024
All
Comparison between totals for 2022, 2023, and 2024 across provinces and occupations.
Sum of _2024_Total Sum of _2023_Total Sum of _2022_Total
Province_territory_and_occu...
Total
Other occupations
Ontario Total
Province/territory n...
British Columbia Total
Quebec Total
Alberta Total
Manitoba Total
0.0M
0.2M
0.4M
0.6M
0.8M
Sum of _2024_Total, Sum of _2023_Total and Sum...
Totals across multiple years and provinces or territories
Sum of _... Sum of _... Sum of _... Sum of _... Sum of _... Sum of _...
Province_territory_and_occupa...
Total
Other occupations
Ontario Total
British Columbia Total
Quebec Total
Alberta Total
Province/territory not...
1122 - Professional o...
Saskatchewan Total
0M
1M
2M
3M
4M
Sum of _2015_Total, Sum of _2016_Total, Sum of _201...